TERM PAPER

On

Data Science in Healthcare

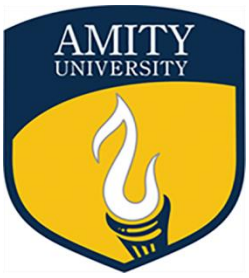
Submitted to

Amity School of Engineering and Technology

Guided By: Ms. Seema Sharma

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AMITY UNIVERSITY UTTAR PRADESH

NOIDA

DECLARATION

I, Kritika Singh, student of B. Tech (2CSE-1-Evening) hereby declare that the project titled "Data Science in Healthcare" submitted to the Department of Computer Science and Engineering, Amity School of Engineering and Technology, Amity University Uttar Pradesh, Noida is an original work done by me under the guidance of Ms. Seema Sharma.

This report is submitted as partial fulfilment for the degree of B. Tech in Computer Science and Engineering and has not been submitted previously to any university.



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CERTIFICATE

This is to certify that Ms. Kritika Singh, student of B. Tech in Computer Science and Engineering has carried out the work presented in the term paper titled "Data Science in Healthcare" as a part of First Year program of B. Tech in Computer Science and Engineering from Amity University Uttar Pradesh, Noida under my supervision.



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ACKNOWLEDGEMENT

The satisfaction that accompanies the successful completion of my project would be incomplete without the mention of people whose ceaseless cooperation made it possible, whose constant guidance and encouragement crown all efforts with success. I sincerely thank all my respected teachers who have helped me with their valuable and appropriate suggestions and supported me throughout the development of my report.

I am highly thankful to my project guide Ms. Seema Sharma who is the biggest driving force behind my successful completion of the project. She has been always there to solve any query of mine and also guided me in the right direction regarding the project. Without her help and inspiration, I would not have been able to complete the project. Lastly, I would like to thank my batch mates who guided me, helped me and gave ideas and motivation at each step.

Kritika Singh

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ABSTRACT

In this digital era, with the movement of an enormous amount of data in the field of healthcare, it is extremely important to plan a proper management system to handle this data flood effectively and efficiently. The objective of this paper is in the same direction. Starting from delivering proper knowledge about the healthcare sector, its definition, the current condition, the need for improvement and the various reasons responsible for degradation in the quality of service provided by the healthcare sector and then taking a turn towards why there is a need for intervention of technologies like Data Science in the field healthcare, how it can enhance the condition of the healthcare sector, pointing the use of Data Science against the COVID-19 Pandemic for explaining how Data Science works in the field of Healthcare the paper mainly focuses on highlighting how Data Science is bridging the gap between technology and healthcare.

Further, the various applications of Data Science in healthcare, for instance, image analysis, drug discovery, predictive analysis, data management etc gives a clear picture of why implementation of Data Science is better than the traditionally used methods. Moreover, the various advantages of using Data Science in Healthcare such as personalized communication, proactive maintenance, timely treatment, advanced care, etc, helps in understanding how Data Science is revolutionizing the healthcare sector. With this one can easily understand that the future of Data Science in Healthcare is going to rise to further heights and hence, Data Science is the key to a bright future because sectors which are incapable of utilizing Data cleverly will not be able to cross the hurdles.

Table of Contents

Declaration……I

Certificate…….....II

Acknowledgement……...III

Abstract……IV

Chapter 1- Introduction……..1

1.1- Introduction…….....1
1.2- Why the healthcare quality is degrading? 1
1.3- What is Data Science? …….....2

Chapter 2- Literature Review……4

Chapter 3- Data Science – Bridging the gap between Technology and Healthcare …….....12

3.1-Need and working of Data Science in the Healthcare sector……12

3.2- Data science applications for COVID-19 Spread……14

Chapter 4- Aarogya Setu- A practical example of application of Data Science for Covid-19 pandemic ...17

4.1- What is Aarogya Setu App? 17

4.2- Role of Aarogya Setu App…….17
4.3- Working of Aarogya Setu App…….18

4.4- How does the app draw Conclusions? …….19

Chapter 5- Applications of Data Science in various fields of Healthcare.. …….21

Chapter 6- Advantages of using Data Science in Healthcare …….26

6.1- Aim of the Healthcare Sector……..26

6.2- Advantages of using Data Science in the Healthcare Sector……...26

6.3- Examples of Data Science in Healthcare ……..26

Chapter 7- Impact of Data Science on Healthcare …….....30

7.1- Impact of Data Science on Healthcare…….30
7.2- How data science is revolutionizing the healthcare sector……..30

Chapter 8- Future of Data Science in the Healthcare Sector ……..31

Chapter 9- Review ……...32

Chapter 10- Conclusion ……...33

References……34

Chapter - 1

INTRODUCTION

1.1.Introduction

Healthcare is one of the salient domains that affect the global population and is linked with the evolution of a country. The healthcare sector can be defined as a sector whose prime purpose is to uplift health. The quality of healthcare service is of utmost importance. This calls for an improvement in the healthcare sector. Further, the healthcare sector being a vital factor in the overall development of any country, raising its benchmarks to further heights is our responsibility.

All the below-listed points are crucial for improving a healthcare system. These include:

1. Service delivery of effectual, safe, quality, personal and non-personal interventions

2. A health personnel that is adequate in numbers competently trained and fairly distributed

3. A health information system that produces, examines, and circulates reliable and prompt information

4. Medical products and technologies that are safe, effective, economical, and accessible

5. A financing system that raises ample funds to ensure that the population can use needed services and is protected from financial catastrophe

6. Governance and oversight of the above

Quality in healthcare is achieved by a supportive relationship between the patient and the healthcare provider. Delivering patients with effective services, fine communication, clear decisions are the signs of good quality service. But due to lack of constant focus on the maintenance of the quality of different sectors, degradation in the quality took place. For example, India is one of the major countries where diseases are still not compliant. The quality of the healthcare sector is dropping at a high rate due to several reasons. It's important to understand what reasons are responsible for this quality degradation.

1.2.Why the healthcare quality is degrading?

There are various reasons observed for the degradation in the quality of the healthcare sector, such as –

1.2.1.Socio-demographic variables

These factors influence the interaction between the service provider and the patient. For example, a physician worked in a health centre in a village but due to the language barrier villagers didn't understand him and didn't obey his prescriptions, which directly affected the quality of service.

1.2.2.Patient’s Uncooperation

Clinical outcomes depend on the cooperation between the service provider and the patient. Improper communication leads to degradation in the quality of service being provided.

For instance, a doctor explained to a patient that this antibiotic medicine might cause an allergy. So, if she had any history of allergies she should let him know. The patient didn’t cooperate

much at the moment which directly impacted the quality of service, as later on the patient came up with a complaint of getting rashes and blamed the doctor for the cause.

1.2.3.Provider Knowledge and skills

Sometimes, inadequate knowledge of the service provider leads to degradation in quality. There are some points such as, empathy, honesty, understanding, commitment, humanity and respect which are essential for a physician to keep in mind to deliver good quality service to their patients.

1.2.4.Immense amount of patients

Prolong working hours, degrades the quality of work. The increasing number of patients demanding medical services does not allow medical staff to work on the quality of medical service. After working for long hours they tend to become lazy and inefficient, which directly affects their way of working.

1.2.5.Resources and facilities

Availability of resources affects the quality of the overall work. There are times when due to unavailability of a patient's proper medical record, leads to treatment failure.

1.2.6.Collaboration and partnership development

Sporadically, doctors don't give the necessary information to patients. For instance, for some particular tests, the patient must not eat a specific food. But due to improper collaboration, the results get affected.

Thus, it can be understood that the quality of healthcare service can improve if we manage to resolve all the above-discussed issues. Along with these reasons, there is one more and the most impactful problem that affects the quality of medical service is a large population, a large population means diversity and a huge amount of data. The health professionals are responsible for a variety of information such as the patient's medical history, clinical records and other medical data. As a vast amount of data moves into clinical practice, new computer-based predictive analytics such as artificial intelligence and natural language–processing algorithms for precision and customized healthcare will help health workers to do their work more effectively. The healthcare data is enormous yet disorganised at the same time. The data points such as X-ray images, nutrient value, etc. of an individual can provide a variety of insights related to his/her current or future health conditions. Such a huge amount of data can be exploited with the help of Data Science, and along with this, all the above-discussed issues can also be resolved.

1.3.What is Data Science?

The field of Data Science has a strong connection with data. The main aim of this field is to extract relevant insights from a given set of data. Data Science has its application in enormous fields. After understanding the literal meaning of the term data science it's important to understand the process involved in the working of data science with a help of pictorial representation.

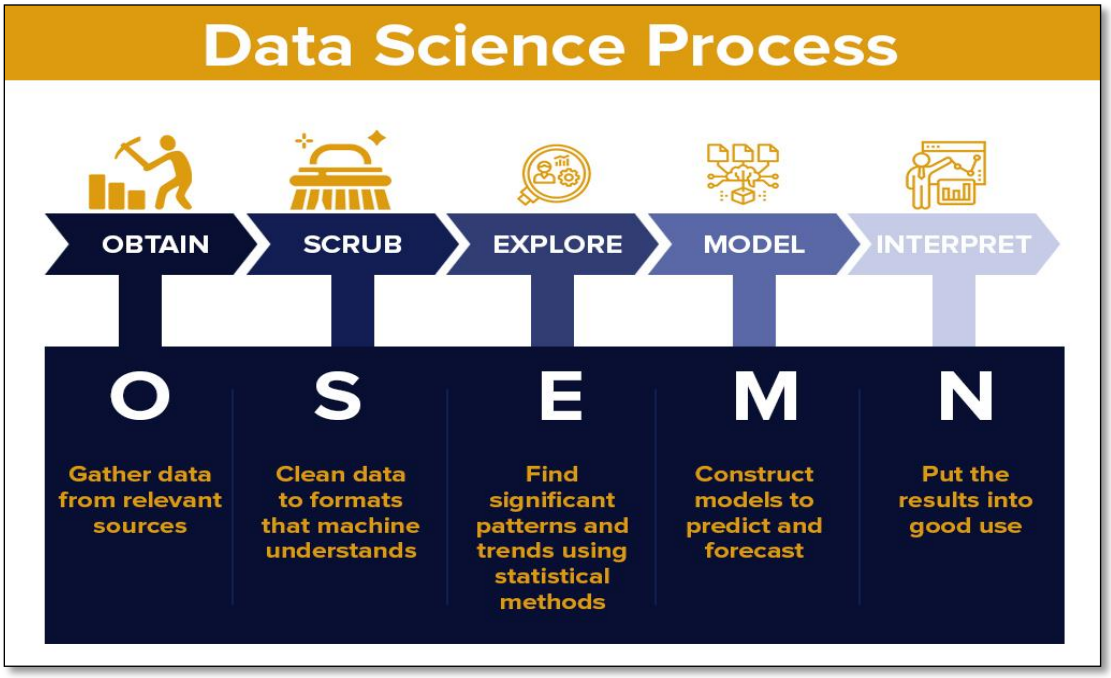


Figure 1.3 Data science process [31]

From all the points discussed above, it can be understood that in order to enhance the cooperation between the service provider and the patients, it's important to improve the quality of the healthcare sector and therefore the concepts of Data Science which helps us in achieving this goal has its importance in the healthcare sector. With this, it becomes important to highlight how data science is bridging the gap between healthcare and technology.

Chapter – 2

LITERATURE REVIEW

S.NO	AUTHOR	YEAR	FOCUS	OUTCOME	TYPE
1	Ali Mohammad Mosadeghrad [1]	2014	The main purpose of this study was to identify all the factors by which healthcare quality gets affected.	The quality depends on various things, such as- the relation between the service provider and the patient, record management, availability of resources, etc and it can be enhanced by effective management and planning.	Journal
2	Richard Lenz ,et al. [2]	2007	The author has focused on linking technology and healthcare.	Technologies reduce the rate of adverse effects by offering precise and prompt information at the time of service.	Paper
3	Ramani KV, et al. [3]	2006	The author has discussed about the condition of health system in India.	By focusing on management, proper education, etc. healthcare quality can be improvised.	Journal
4	Y Balarajan, et al. [4]	2011	The author has discussed about the hurdles to equity in service being provided in India.	Improper distribution of resources, high expenditures, poverty, are all responsible for inequality in health service .	Paper

5	Dr. Liji Thomas [5]	2021	The author has discussed about healthcare systems in different countries and their response to COVID-19 Pandemic.	By looking at how different countries responded to the COVID-19 Pandemic it can be concluded that no country has a best healthcare system, but it can be achieved by focusing on factors such as public funding, utilisation of technology, resource distribution and health monitoring.	Article
6	Mark A. Schuster, et al. [6]	1998	The author has discussed about the healthcare quality of the United States.	Good healthcare quality means providing patients with adequate services, fine communication, accurate decision making and cultural respect.	Paper
7	Hian Chye Koh, et al. [7]	2011	The author has discussed about the applications of data mining in health.	The enormous amounts of data produced by healthcare sectors are complex to handle via traditional methods. By using data mining , this vast amount of data can be managed effectively and efficiently and also valuable insights can be drawn from the data and relevant predictions can be made.	Article
8	Longbing Cao [8]	2017	The author gave a survey on Data Science.	We are facing a highly evolving data universe. Utilizing the concepts of Data Science in different fields is seems to be very important.	Article

9	Aishwarya Kumar, et al. [9]	2020	The author has discussed about fighting against the COVID-19 Pandemic by making use of various technologies.	From image analysis to making predictions, technologies has helped standing against the COVID-19 Pandemic.	Paper
10	Hyunghoon Cho, et al. [10]	2020	The author has discussed about the Privacy issues and contact tracking apps.	The privacy related issues related to the contact tracking apps can be resolved by using some techniques like location logging etc, which helps managing the privacy along with using the application.	Paper
11	Michael Ashburner, et al. [11]	2000	The author has discussed about the Genomics and genes.	Genomics is the branch of biology focusing on the structure, evolution and mapping of genomes. A genome is comprised of the DNA and all the genes of the organisms.	Article
12	RD Hawkins, et al. [12]	2010	The author has discussed about the genomics and the future of genomics.	Data integration can be used to identify elements in the genome, explore the function of gene.	Article

13	David B. Searls [13]	2005	The author has discussed about data integration and drug discovery.	The integration of data will be important to future drug discovery.	Article
14	Vasant Dhar [14]	2012	The author has discussed about Data Science and predictive analysis.	Model building and decision making can be very fruitful for predictive analysis.	Paper
15	Jinchuan Chen, et al. [15]	2013	The author has discussed about the challenges faced in data management of big data.	Today there is a vast amount of data present, testing the data management system. Challenges offered by this enormous amount of data are in three ways-volume, velocity and variety.	Article
16	Anne E Carpenter, et al. [16]	2006	The author has discussed about image analysis software.	Image analyss can be used for detecting some things which cannot be detected by naked eyes. For instance, texture of DNA.	Paper

17	Atam P. Dhawan [17]	2011	The author has discussed about medical image and medical image analysis.	Based on image analysis, doctors can take relevant and meaningful decisions for their patients.	Book
18	Robby Gupta [18]	2019	The author has discussed about the cases where data science is greatly impacting the healthcare.	Drug detection, Wearable, Diagnostics, Public Health, etc are some of the areas that are highly impacted by Data Science.	Article
19	Sharvari Raut [19]	2021	The author has discussed about the Data Science applications.	Data Science can be used in many areas. For instance, medical image analysis, virtual assistant, drug discovery, and many more.	Article
20	Sandra Durcevic [20]	2020	The author has discussed about big data and its various applications that can improve health process and can save lives.	By real time alerting, electronic health records, strategic planning, enhancement in security, telemedicine, smart staffing, etc. Data Analytics can improve and save lives.	Article

21	Shailendra Sinhasane [21]	2019	The author discussed about the different healthcare areas that got benefited by the application of Data Science.	Data Science has improved healthcare sector in many ways .Improved diagnostic accuracy, medicine discovery in short time span, improved management, early diagnosis of deadly diseases, etc.	Article
22	Luigi Atzori, et al. [22]	2010	The author has discussed about the IoT-internet of things.	IoT enables interaction between smart tools and humans. Vast amount of data can be obtained easily anywhere and anytime.	Paper
23	Sapna Tyagi, et al. [23]	2016	The author has discussed about applications of Internet of things and the role it plays in healthcare.	IoT enables healthcare systems to focus their efforts on clinically relevant services and improved patient outcomes which will make health monitoring more effective and efficient.	Paper
24	Geoff Appelboom, et al. [24]	2014	The author has discussed about the clinical utilization of wearable.	In healthcare, wearable tools are very useful. These tools enable self-monitoring, tracking, and medical data, for healthcare providers.	Paper

25	Shivayogi Hiemath, et al. [25]	2014	The author has discussed about the application and function of wearable tools.	Wearable sensors enable effective communication between doctors and patients, enabling digital discussions, feedback, and new treatments.	Paper
26	Jishnu Das, et al. [26]	2012	The author has discussed about the quality of services provided to rural and urban people in India by the healthcare sectors.	Improper diagnosis and incorrect treatments, were widely spread. Poor maintenance of healthcare was observed all over.	Article
27	Sabyasachi Dash, et al. [27]	2019	The author has focused on huge healthcare data and its impact on healthcare.	Data will raise the level of healthcare to further heights. From effective drug discovery to making relevant insights from data, it will improve the healthcare sector. It is believed that application of big data analytic in healthcare might lead to lot of cost reduction.	Journal

28	Leah Davidson [28]	2019	The author has discussed about how Data Science brought revolution in healthcare	Data science can save lives by predicting the chances of occurrence of a particular disease, providing medical help, customizing treatments and finding early cures for deadly diseases.	Article
29	Jessica Kent [29]	2020	The author has discussed how models can impact the healthcare decisions.	From providing guidelines to suggesting preventive measures during The on-going COVID-19 Pandemic is a proof of how Data Science is improving the decision -making in the health sector.	Article
30	W Van Der Aalst [30]	2016	The author has briefly discussed about process drilling and importance of Data Science.	Process drilling bridges the gap between data mining and management. Further, Data science is the basic requirement for future, because organizations that are unable to use data in a cleverly will not be able to cross the hurdles.	Article

Chapter - 3

DATA SCIENCE - BRIDGING THE GAP BETWEEN TECHNOLOGY AND HEALTHCARE

3.1. Need and working of Data Science in the Healthcare sector

Why do we need data science in the healthcare sector? How does data science works in the healthcare system? This chapter provides answers to these queries.

3.1.1.Why do we need data science in the healthcare sector?

Today, we are completely covered with lots of data. According to a study, the data generated by every human each day is enormous. This data includes heart rate, stress level, sugar level, and many more. To handle such an enormous amount of data, we have more advanced technologies and Data Science is one of them. Systematic data management, analysis, and interpretation of data can revolutionize the healthcare sector by enhancing the quality of service being provided. This is the reason why different industries, including the healthcare industry, are taking strong steps to convert these potential datasets into better services and financial benefits.

From the field of genomics to predictive analysis, several areas in the healthcare sector that require the application of Data Science to enhance the quality of work being done.

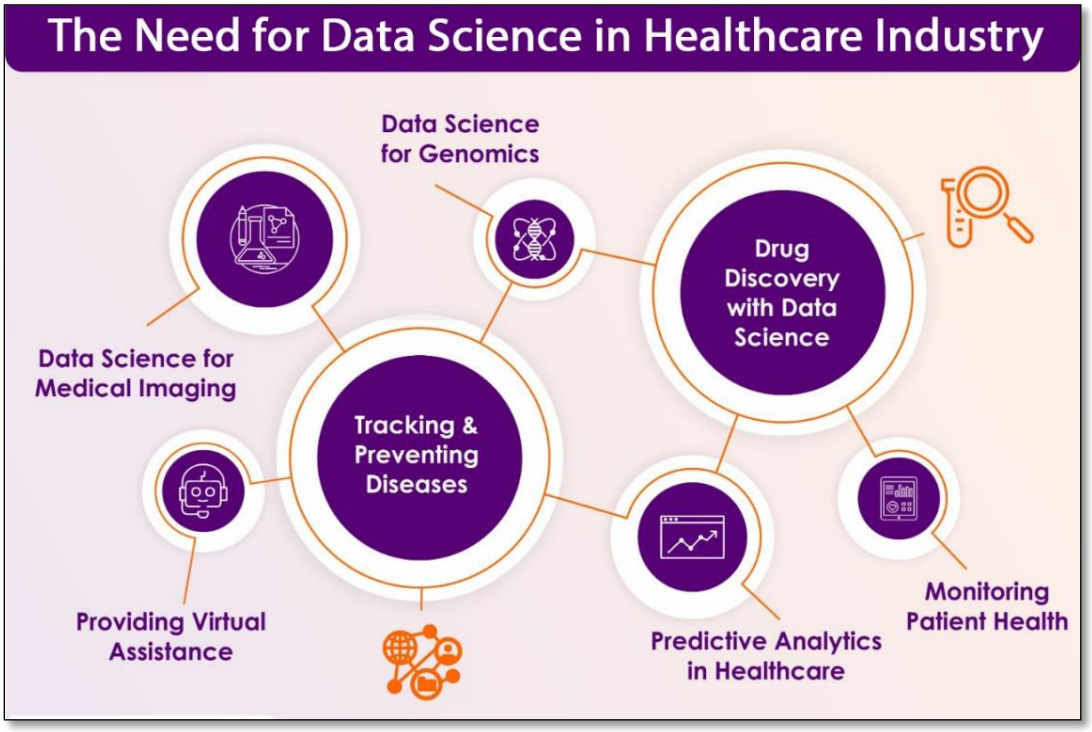


Figure 3.1.1 Need of data science in healthcare industry[32]

3.1.2.How does data science works in the healthcare sector?

In the Healthcare sector, Data Science has a variety of tasks to perform. From identifying high-risk patients to optimizing the network for good quality care, Data Science has a variety of work to do. With time, Data Science has outperformed itself and its applications have practically stretched to all spectrums of everyone's lives. Amidst the COVID-19 scare, data science applications are our guiding beacon. How exactly?

The COVID-19 pandemic has raised an alarm to reconsider the existing healthcare system and has forced people to come up with better solutions that can cater for the healthcare needs of people despite the ever-increasing population. The traditional healthcare system that is not supported by technology has proved to be unstructured to large extent. Data science is playing a commendable central role in the global response to the COVID-19 pandemic.

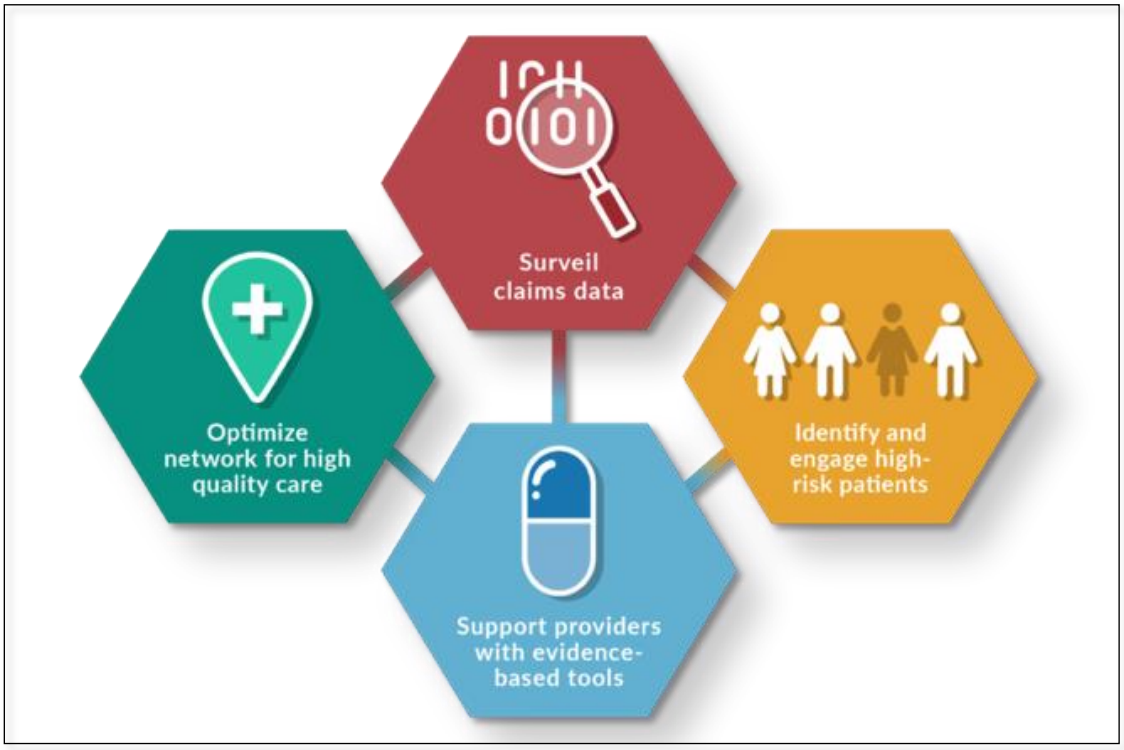


Figure 3.1.2 Working of data science in healthcare[33]

3.2.Data science applications for COVID-19 Spread:

The ongoing COVID-19 pandemic situation which we all are fighting is the best example for understanding how Data Science is working in the field of healthcare.

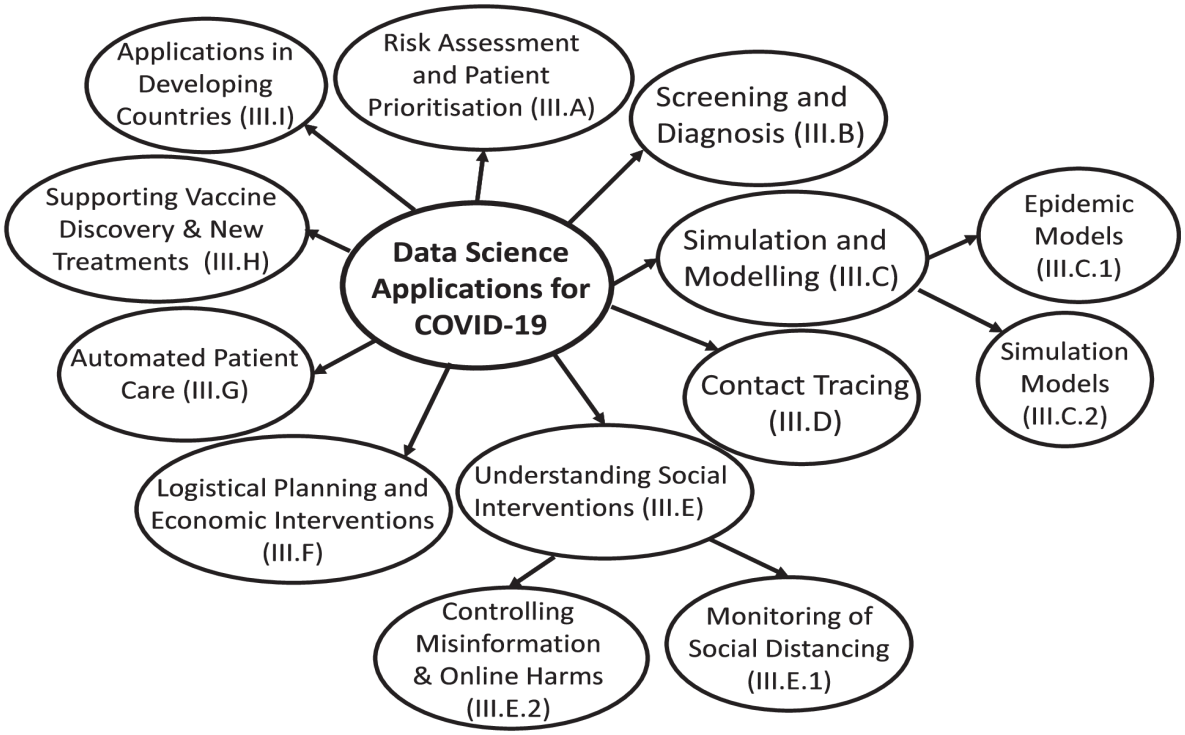


Figure 3.2 Data science applications for COVID-19[34]

3.2.1.Risk Assessment

The Healthcare sector is facing unprecedented stress on its resources (e.g., availability of intensive care beds, availability of oxygen cylinders). This calls for a speedy assessment and management of patient's risk while allotting resources judiciously. At the time of peak load, this must be done promptly and precisely. Due to diverse symptoms and conditions utilizing technologies for data-driven risk assessment and management in discrete COVID-19 patients would be useful. Attributes such as age, gender, or health condition can be used to give an estimation of mortality risk rate.

3.2.2.Screening and Diagnosis

Countries with increasing COVID-19 infection rates are facing a great issue related to the lack of proper screening and diagnosis facilities. It hinders capacity management and social distancing norms since asymptomatic people, are often unconscious that they are the carrier of the virus. Implementation of Data Science concepts helps in managing this.

3.2.3.Simulations and Models

Here are the details of how models and simulations can help reduce the impact of COVID-19.

3.2.3.1.Epidemic Models

In epidemiology, there is the broad usage of compartmental models. These models comprise different compartments in which populations are divided and the flow of people among compartments is modelled using the mathematical concept of ODE (ordinary differential equation). For instance, the SEIR model is used for COVID-19's spread which models the movement of people between four compartments: S-susceptible, E-exposed, I-infected, and R-recovered. This helps in maintaining records systematically and effectively.

Parameterising the above models requires up-to-date information on the virus spread.

3.2.3.2.Simulation Models

Pandemic generates a large number of questions all of which cannot be answered via the above model only. For example, taking decisions related to quarantine and social distancing strategies, resource management plan, the capacity of hospital beds, intensive care units, staffing, and care decisions like deciding for admission and discharge of patients. Hence, simulation models help in providing relevant decisions on the above points.

3.2.4.Contact Tracking

Most countries countered to the early stages of COVID-19 with basic containment measures. This involves promptly inspecting infected people, putting them under quarantine and further tracing the contacts. But this needs to take place more rapidly and effectively therefore techniques such as mobile contact sensing, online surveys and automated diagnosis have all been proposed to swiftly identify exposure.

3.2.5.Understanding Social Interference

Governments have taken relevant steps to stand against the COVID-19 spread.

3.2.5.1.Observation of Social Distancing

Many countries have implemented social distancing norms to alleviate the spread of COVID-19. This is a non-pharmaceutical intervention implemented to avoid any kind of contact among people and therefore reduce the spread of the COVID-19 virus. Data science enables contact tracing for the monitoring of social distancing.

3.2.5.2.Controlling the spread of false belief

The spread of false information has threatening repercussions. For example, that vinegar is more effective than hand sanitiser against COVID-19. Naturally, people who trust such misinformation could risk their lives to great extent.

3.2.6.Logistical Planning and Economic Interventions

COVID-19 has also impacted healthcare supply chains. For instance, the supply of protective equipment such as masks and PPE kits, intensive care tools such as test kits, beds, and ventilators gets greatly affected. Application of data science techniques enables judicious allocation of resources. Further, social distancing norms have an immense impact on the global economy. As companies come out from economic hibernation it’s a difficult task to return to normal levels of service and operation. Data scientists might be able to assist in detecting the problems and offer

relevant solutions. For example, companies can use data science to determine odd patterns of behaviour in the market.

3.2.7. Automated Care

Due to this COVID-19 outbreak, there is a shortage of healthcare staff. To overcome this, self-operating tools could be developed. Such systems can help in providing information about the virus and can suggest precautionary measures.

For example, remote care tools like oximeters enable the detection of oxygen levels so that a user can check the severity of the infection.

3.2.8. Aid to Vaccine Discovery and New Therapies

The global effort to discover drug treatments and vaccines can also benefit from data science. For instance, many computational methods can reduce the time spent on analyzing data and also helps in identifying eligible patients for clinical trials, which is often a time-taking process.

3.2.9. Applications in Developing Countries

To control the worsening of the current pandemic situation, countries need to act more rapidly and effectively. Making use of Data Science for contact tracing, remote treatments, etc, helps to fight against the pandemic with more efficiency and effectiveness.

From the above discussions, it's now clear how data science is bridging the gap between technology and healthcare. But just acquiring theoretical knowledge is not enough and thus it calls for a demonstration of the practical application of data science in healthcare which is what the next chapter is about.

Chapter – 4

AAROGYA SETU- A PRACTICAL EXAMPLE OF APPLICATION OF DATA SCIENCE FOR COVID-19 PANDEMIC

4.1.What is Aarogya Setu App?

Today, we all are fighting this deadly coronavirus and every individual must act responsibly and sensibly. For the time being following social distancing and opting self –isolation are the best we can do. Aarogya Setu App is a great step given the COVID-19 virus. From providing self-assessment tools to giving information about the vaccines, this app has proved to be a helping hand during this dreadful situation.

Aarogya Setu App (Shield for a safer India)



Figure 4.1) Aarogya Setu- Application of data science for COVID-19[35]

This is a mobile application aiming to sensify the citizens of the country in a joint battle against the COVID-19 virus. The app encourages the idea of precaution, self-isolation, and social spacing. It uses an active Bluetooth connection to find out if you have been near a COVID-positive person. It includes several Data Science concepts to examine COVID-19 spread in India.

4.2.Role of Aarogya Setu App

As the virus spreads via being near to an infected person, taking immediate precautions or avoiding contact with an infected person in the first place are the solutions. But if there is any case of contact, the app suggests taking precautions and opting for quarantine. This is done so as to stop the infection from spreading. Moreover, if a person tests positive, a notification will be sent to people who've been in close contact with that person in the past 28 days based on their anonymous IDs. Anyhow the identity of the COVID-positive individual is not going to be

disclosed. The whole process is complex and needs data analysis of raw data to make conclusions.

4.3. Working of Aarogya Setu App

1. The app starts by asking for your contact details for signing up. It then delivers you a privacy notice to notify you that the app will collect and use certain data.

2. It will then appeal access to your device location following which you need to switch your Bluetooth on for 120 seconds.

3. For the app to work effectively, constant availability of Bluetooth and GPS is required in order to collect data and analyse it to draw relevant insights.

4. Moreover, the app requires additional details such as full name, gender, age, travel history of the last 30 days, etc.

5. The app entails a self-assessment questionnaire wherein it asks: Gender and age, symptoms or any other issue. History of international travel and questions pertaining to recent interactions with COVID-positive person are also asked.

6. The app utilizes the acquired information and place an individual under a low risk or high-risk class.

7. The app assures that your data will only be shared with the Government of India in an anonymised form causing no risk to your privacy.



Figure 4.3) Working of Aarogya Setu App[36]

After understanding the working of the Aarogya setu app in detail the next step is to understand what data it leverages to conclude?

4.4. How does the app draw Conclusions?

1. For dropping conclusions, the app keeps a constant track of all the location data where you came in contact with the other app user. Further by going through the collected information, it informs users about any risk of coronavirus infection around them.

2. The personal information entered by the user is used in an anonymised form by the government and is further used for COVID-19 management by creating reports.

3. The self-testing feature allows the user to assess whether he/she is suffering from coronavirus or not. Then as a conclusion, the app drops a report based on the input data filled by the user during the assessment.

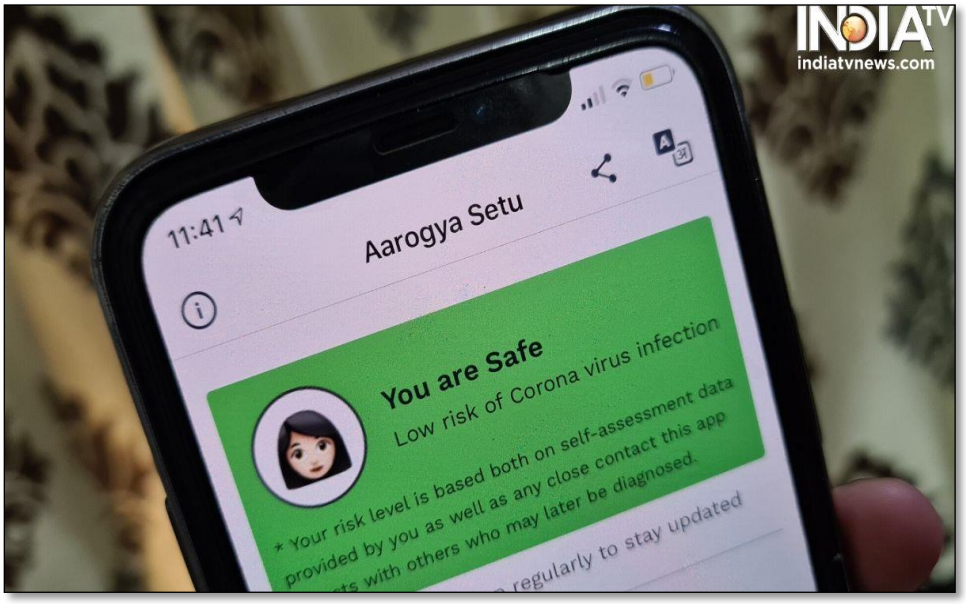


Figure 4.4) How the Aarogya Setu app draws conclusions[37]

From genomics to image analysis, Data Science has a broad range of applications in the field of healthcare. The upcoming chapter is going to give a brief about the same.

Chapter - 5

APPLICATIONS OF DATA SCIENCE IN HEALTHCARE

5.1. Application of Data Science in various fields of healthcare

Data Science is rapidly budding to inhabit all the industries of the world today. There are enormous fields in healthcare like image analysis, drug discovery, genomics, predictive analysis and more that make use of data science.

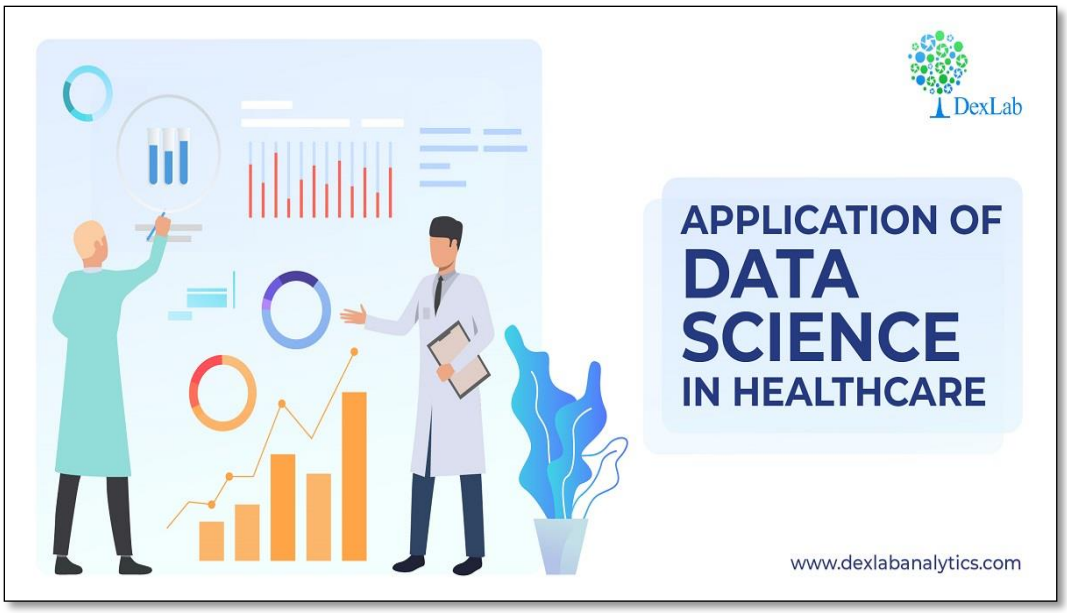


Figure 5.1) Applications of data science in healthcare[38]

The various fields of healthcare where Data Science is applicable are as follows:



Figure 5.2) Fields where data science can be used[39]

5.1.1.Data Science for Genomics

Genomics is the branch of biology focusing on the structure, evolution and mapping of genomes. A genome is comprised of the DNA and all the genes of the organisms. A gene has three aspects- biological, molecular, and cellular. Before the availability of powerful computation, organizations expended a great amount of time and money on analyzing the order of genes. This was an expensive and monotonous process. However, with advanced data science tools and concepts like data integration, it is now practicable to analyse and procure insights from the human gene in a much shorter time and in a much inexpensive manner. The main aim of the researchers is to analyse the genomic strands and look for irregularities and defects in them. Then, they find links between genetics and the health of the person.

Moreover, a study of genomics also includes finding the right drug which comes up with a deeper insight into the way a drug reacts to a particular genetic issue.



Figure 5.1.1) Data science for genomics[40]

5.1.2.Data Science for Drug Discovery

Drug Discovery is an extremely complex discipline. Pharmaceutical industries are greatly dependent on data science to solve their issues and produce finer drugs for the people. Drug Discovery is a gradual process that calls for heavy financial expenditure and testing. Data Science and Machine Learning algorithms are transfiguring this process and providing considerable insights into optimizing and increasing the success rate of predictions. Also, deep learning algorithms can find the expectation of the development of disease in the human system. The data science algorithms enable us to replicate how the drugs will function in the human body taking away the lengthy laboratory trial and error. With the evolution in the data-science facilitated drug discovery, it is now attainable to ameliorate the collection of historical data to assist in the drug development process.

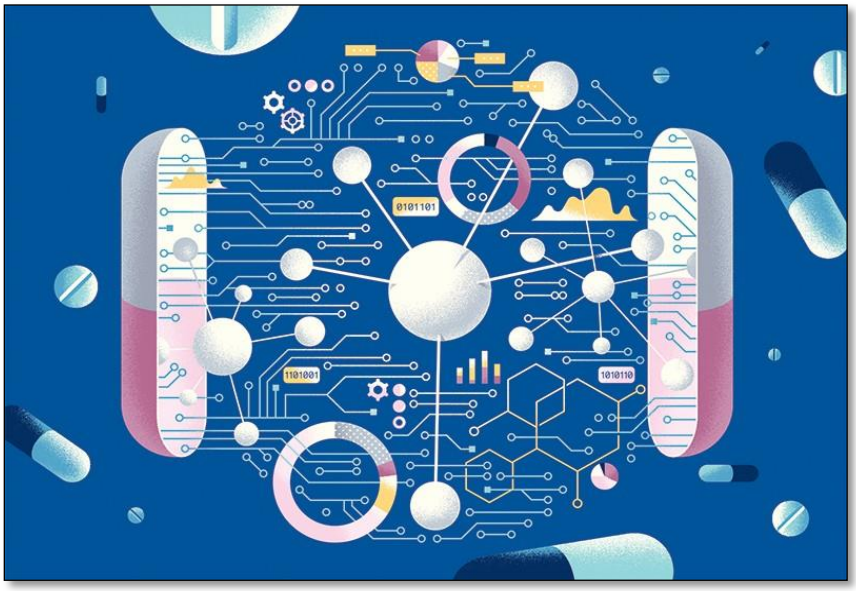


Figure 5.1.2) Data science for drug discovery[41]

5.1.3.Data Science for Predictive Analysis

Healthcare is a prime domain for predictive analytics. A predictive model utilizes factual data, learns from it, discovers patterns and generates precise prophecies from it. It observes numerous connection and association of symptoms, discover habits, diseases and then construct relevant predictions. Predictive Analytics has an important role in ameliorating patient's care, chronic disease governance and increasing the efficiency of supply chains and pharmaceutical logistics. It is a data-driven approach focusing on the anticipation of diseases that are often prevalent in society. With data science, hospitals can anticipate the worsening in patient's health and furnish the preventive measures and begin an early therapy that will aid in lessening the possibility of the further aggravation of patient's health.

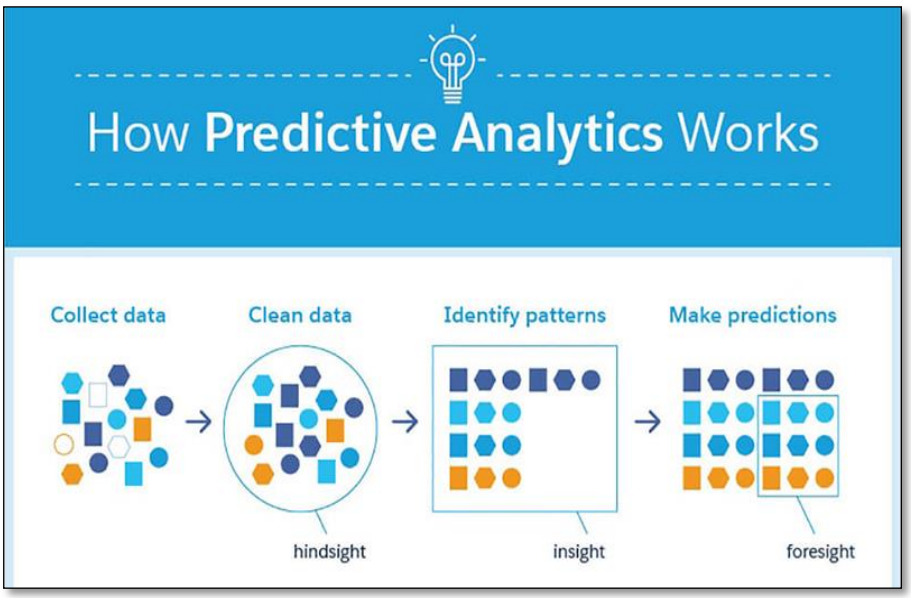


Figure 5.1.3) Data science for predictive analysis[42]

5.1.4.Data Science for Data Management

Considering that a vast quantity of data is unmanageable via utilizing conventional tools, technically advanced applications and tools are required for such a task. Data science works with a variety of domains including data management and analysis, to withdraw deeper insights for enhancing the functionality or performance of a healthcare system. Moreover, with the availability of some of the most innovative and relevant ways to visualize large data post-analysis, it has become feasible to understand the functioning of any complex system.

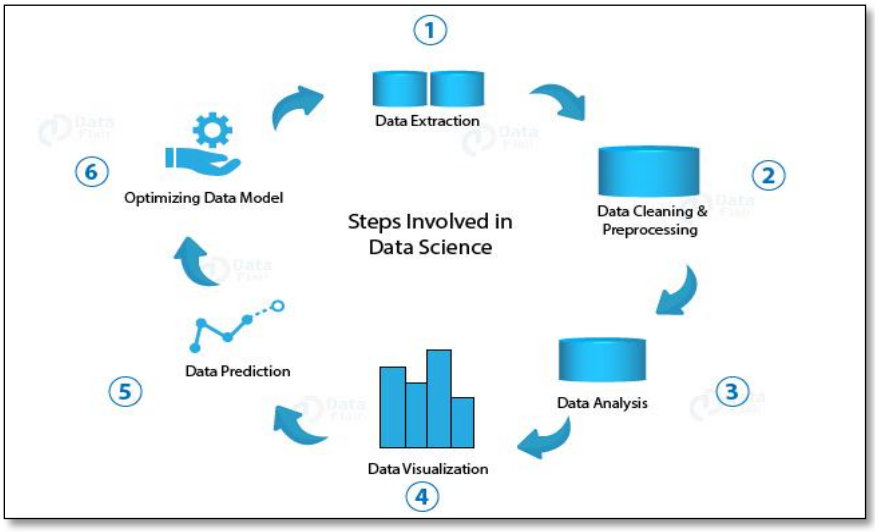


Figure 5.1.4) Data science for Data Management[43]

5.1.5.Data Science for Virtual Assistance

Using disease predictive modelling, an exhaustive virtual platform have been established by data scientists that assist the patients. Via these platforms, a patient can fill up his or her symptoms as the input and get insights about the different possible diseases based on the confidence rate.

Furthermore, patients who suffer from psychological issues like depression, anxiety, and stress etc, can use these platforms to get help in performing their daily duties.



Figure 5.1.5) Data science for Virtual Assistance[44]

5.1.6.Data Science for Image Analysis

The prime and principal use of data science in the health industry is via image analysis. There are various medical imaging techniques like X-Ray, MRI and CT Scan that envision the inner parts of the human body. Traditionally, doctors would manually inspect these images and search asymmetry within them. But, it was often hard to find minute deformities and as a consequence, doctors could not suggest a proper diagnosis. But now with data science, it is feasible to observe such microscopic malformations in the scanned images. Through image dissolution, it is possible to observe defects present in the scanned images. Advancements in medical image analysis have made it possible for doctors to find out microscopic tumours that were otherwise hard to find and also helped them in taking important decisions.

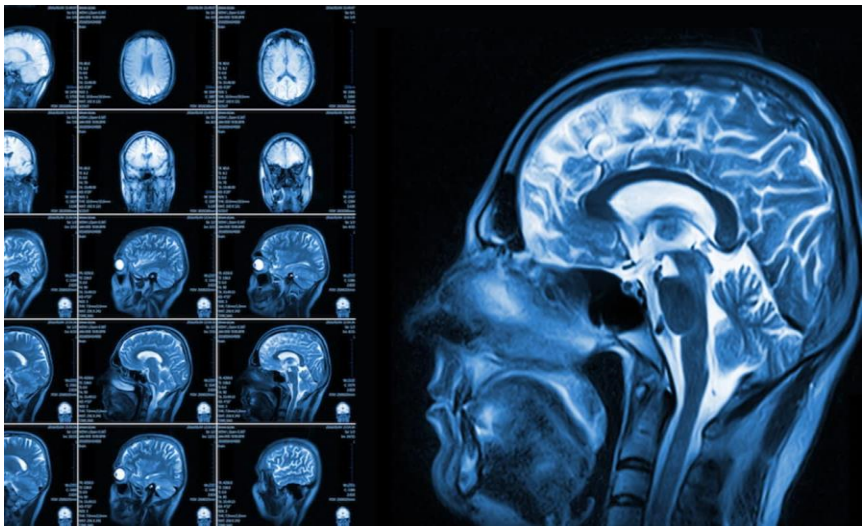


Figure 5.1.6) Data science for Image Analysis[45]

Along with the application side of Data Science in Healthcare, the advantages of using Data Science in the Healthcare sector is equally important to discuss. The upcoming chapter is going to explain about the same.

Chapter - 6
ADVANTAGES OF USING DATA SCIENCE IN HEALTHCARE

6.1. Aim of the Healthcare Sector

The main goal of the healthcare sector to enhance the quality of service been provided is as follows

1. To alleviate the functionality of the healthcare system
2. To lessen the possibility of treatment failure
3. To supply genuine treatment on time
4. To evade dispensable emergency due to the unavailability of doctors
5. To maintain equity in healthcare

With the application of data science, all the above goals are achievable.

6.2. Advantages of Data Science in the Healthcare

Below listed are some of the top advantages of data science in healthcare-

1. Improved Functioning Efficiency
2. Advance Care and Therapy
3. Diagnosis of the Accurate Cure for Diseases
4. Personalize Communication.
5. Enhanced Access to Vital Information.
6. Proactive Maintenance of Equipment.

6.3. Examples of Data Science in Healthcare

6.3.1. IoT (Internet of things)

IoT is growing immensely around the world and especially in the field of healthcare. IoT devices create a continuous flow of data while tracking the health of people which make these devices a vital donator to big data in healthcare. Such resources can connect several devices to provide solid, effectual and good healthcare service. By using IoT devices efficiently, a doctor can measure and track several parameters from the clients in their specific locations either, home or office. Therefore, via untimely intervention and therapy, a patient might not require hospitalization or even visit the doctor resulting in a significant cost reduction in healthcare expenses.

Examples of IoT devices include health-tracking biosensors, clinical devices, etc.

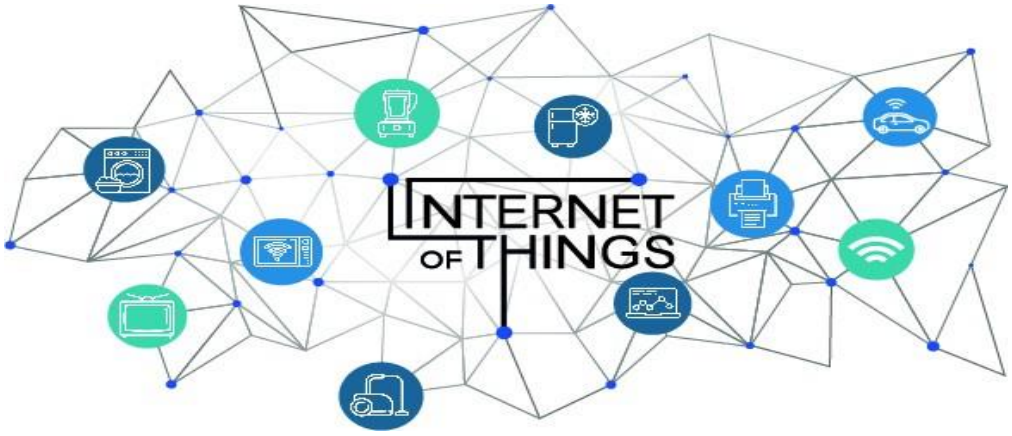


Figure 6.3.1) Internet of things[46]

6.3.2.Wearables

These are rapidly becoming pervasive in the healthcare sector. Aside from being a cool accessory, they encourage self-health governance among people. The device records vital health details like blood pressure, heart rate, sleep pattern, pulse etc. Using the concepts of AI, machine learning, etc, data scientists can utilize this information from wearables to deduce relevant insights. Via advanced analytical models, variations in patient's health can be determined, and any disorder or a concerning symptom can be detected. This grants doctors to predict a health issue and supply preventive care.

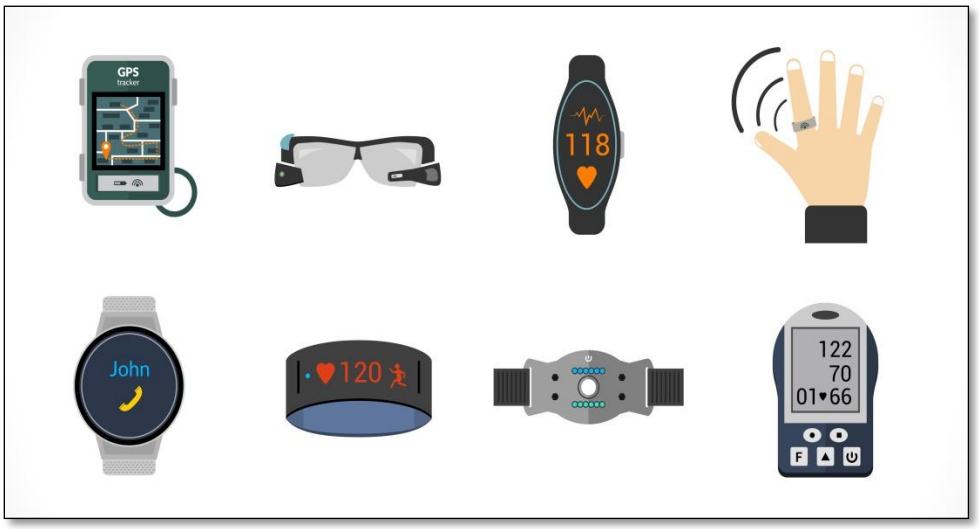


Figure 6.3.2) Wearables[47]

6.3.3.Diagnostics

Diagnosis is a crucial part of healthcare and a proper diagnosis is rare. But, by using data science, precise diagnosis can be achieved. From predicting the probability of disease to providing an early cure, Data Science can be of great use.



Figure 6.3.3) Diagnostics[48]

6.3.4.Public Health

Numerous healthcare organizations have already started to grasp data from different sources to improve general public health. This data has the key to understanding the overall public health in a specific region and the signs of a forthcoming health crisis in that region can also be understood. Moreover, through this information, the availability of medical aids can also be checked in that specific region.

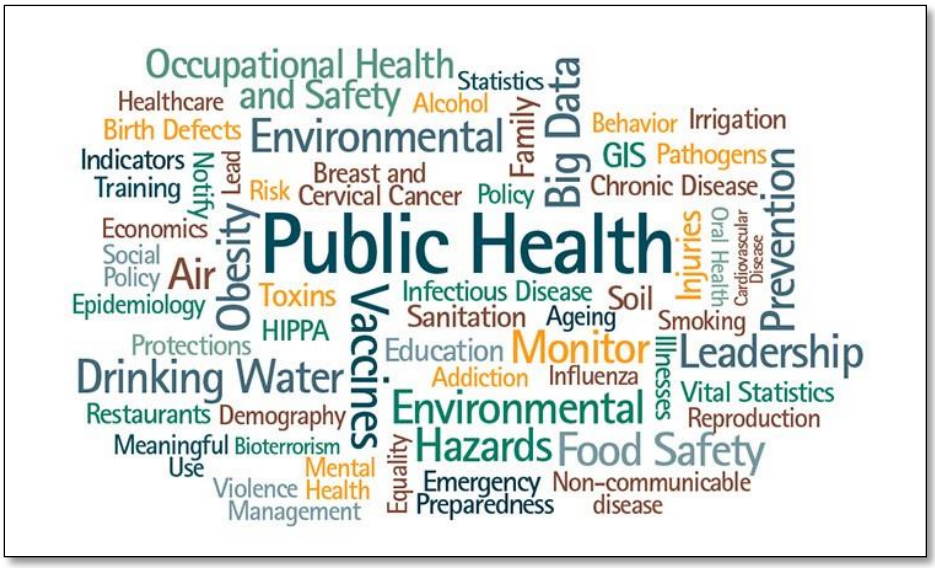


Figure 6.3.4) Public Health[49]

6.3.5.Reduced Health Cost

Healthcare expenditure appears to be rising with time which has a key impact in delivering a superior patient experience. By the application of data science, details about the billing data and clinical system details can be checked. This allows them to dig into the trends in room usage and required resources available to serve patient needs; thus, helping identify potential areas of operational gaps and revenue losses.

These data-driven provisions can create a way for reduced operational costs which translate to lower healthcare expenditure for patients with better satisfaction.



Figure 6.3.5) Reduced Health Cost[50]

6.3.6.Optimal Staffing

Healthcare needs are only going to rise, and providers may often find it difficult to have adequate medical staff for patient care at any given point in time. Any change in the patient stream will affect the units working. More than required staff will increase labour expenditure, while, an insufficient amount of staff can end up working overtime and reach an exhaustion stage.

So, how to keep the optimum staff available? Data Science has the solution.

It can predict alterations in inpatient visits based on records over the years and come up with a pattern in staff allocation based on admission rates in the past. This would provide an idea as to when the centre is going to need what staff level. Moreover, all required resources can be allocated to patients suitably and timely.



Figure 6.3.6) Optimal Staffing[51]

The most important point which still remains untouched is how Data Science is revolutionizing the healthcare sector which means how Data Science has impacted the healthcare.

Chapter - 7
IMPACT OF DATA SCIENCE ON HEALTHCARE

7.1. Impact of Data Science on Healthcare

Usage of Data Science has impacted the healthcare sector in many ways. From personalized medicines to improved monitoring, Data Science has proved to be beneficial for the healthcare sector. Also, with the advent of abundant innovative tools and technologies, doctors are in a position to monitor patient's conditions from remote locations. In earlier days, doctors and hospital management were not able to hold multiple numbers of patients at the same time. And due to the absence of proper treatment, the patient's condition used to get worse. However, now, the scenario has changed. In today's time with the use of wearable devices which are based on the concept of Data Science and Machine Learning, doctors can be notified about the health condition of the patient. Then, hospital management can send their subordinate doctors, assistants, or nurses to these patient's homes. Doctors get this real-time data of the health through updates and notifications in mobile applications, they can then diagnose the condition. This is the way Data Science helps in caring for patients via technology.

7.2.How data science is revolutionizing the healthcare

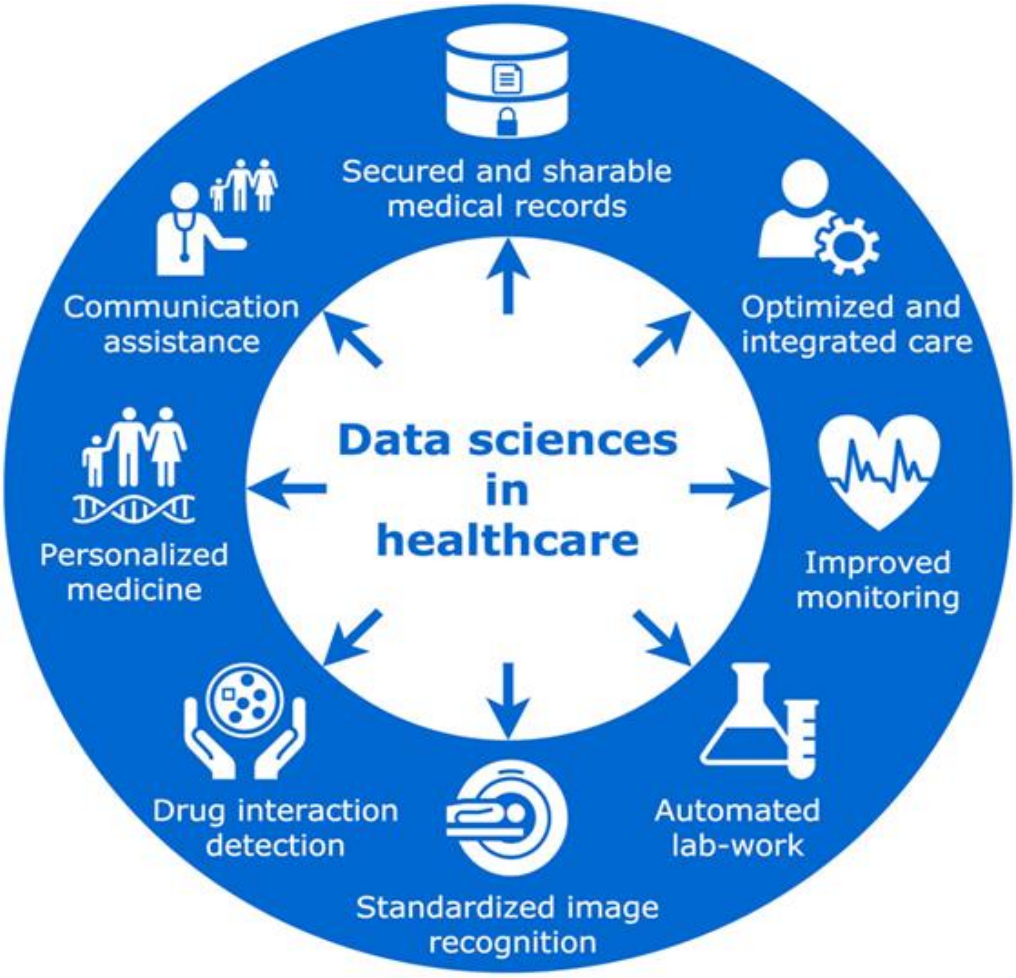


Figure 7.2) How data science is revolutionizing healthcare sector[52]

Chapter - 8

FUTURE OF DATA SCIENCE IN THE HEALTHCARE SECTOR

8.1. Future of Data Science in the Healthcare Sector

In this digital era, the data that we diligently collect will finally pay us back. Today when quality is the prime focus, applications of Data Science in various fields will ensure the service being provided is of the utmost quality. Talking about its application in the healthcare sector, it offers various facilities such as forecasting the chances of disease, offering medical guidance, personalizing therapies, and finding an early heal for diseases. With several interactive tools such as Fitbit, sooner or later the prime source of data will be our houses and the devices encircling us. As Data Science is already doing wonders, there is no confusion that its application will show itself to be more indispensable in the coming future.



Figure 8.1) Future of data science in healthcare sector[53]

Data Science will raise the level of the healthcare sector to new heights. Doctors will get adequate assistance and patients will get a more customized experience and perfect treatments. Hence, Data Science is the key to a better future for all the different sectors, because sectors which are incapable of utilizing data in a cleverly will not be able to cross the hurdles.

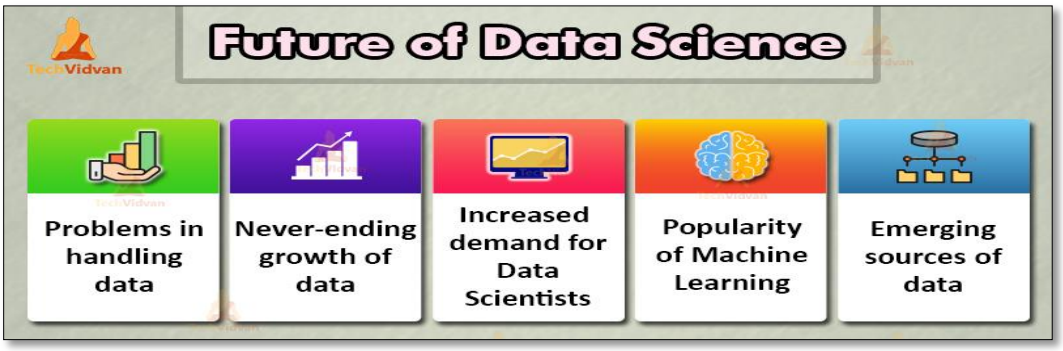


Figure 8.2) Future of data science[54]

Chapter - 9
REVIEW

9.1 Review

By the means of this paper, it can be understood that the main aim of the healthcare sector is to maintain health and offer adequate treatment. But due to several reasons such as resource unavailability, the vast amount of patients, etc, the quality of service is degrading. Here comes the need for technologies such as Data Science which is a field that is based on extracting meaningful insights from the available data, so that healthcare service quality can be improved. The utilization of Data Science to fight against the COVID-19 Pandemic is one of the best practical applications of Data Science in the field of Healthcare. Further, the various application of Data Science in the healthcare such as drug discovery, genomics, etc and the plenty of advantages that can be achieved by the healthcare sector after the implementation of Data Science has highlighted how Data Science has transformed the Healthcare sector.



Figure 9.1) Data Science in Healthcare[55]

Chapter - 10

CONCLUSION

10.1 Conclusion

In order to conclude, it can be said that the combination of Data Science and Healthcare is doing wonders. From resolving various health issues which were difficult to tackle via traditional methods to providing relevant insights from the huge healthcare data, Data Science has out-performed itself and has commendably connected technology and healthcare. As the amount of data is undoubtedly going to increase with each passing day, the significance of Data Science in the field of Healthcare is never going to get vanished. There is a great scope of Data Science in the Healthcare sector from medical image analysis to providing an early cure for diseases, the opportunities for one who has an interest in working with Data Science in the Healthcare sector is immense. Further, it can be said that the implementation of Data Science in the Healthcare Sector is a wonderful portrait of the bond between the technical and the medical field.

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